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Numerical Methods

Lab 5

Newton Method and Aitken Acceleration

1-Purpose of the Lab

The purpose of this lab is to examine Newton's method in two different situations. First, Newton's method is used again for the function

$$f(x) = x\sin(4x)$$

in order to see the effect of the starting value. Since this function has several roots, different initial guesses can produce different final roots.

After that, Newton's method is applied to a nonlinear system with three variables. In this part, the function vector and the Jacobian matrix are constructed. Lastly, Aitken's acceleration is applied to a one-dimensional iteration to see how it improves convergence.

2-Effect of the Initial Guess

Newton's method does not always give the same root when the function has multiple roots. For the function

$$f(x) = x\sin(4x)$$

there are many points where the function crosses the x-axis. Because of this, the selected initial value plays an important role.

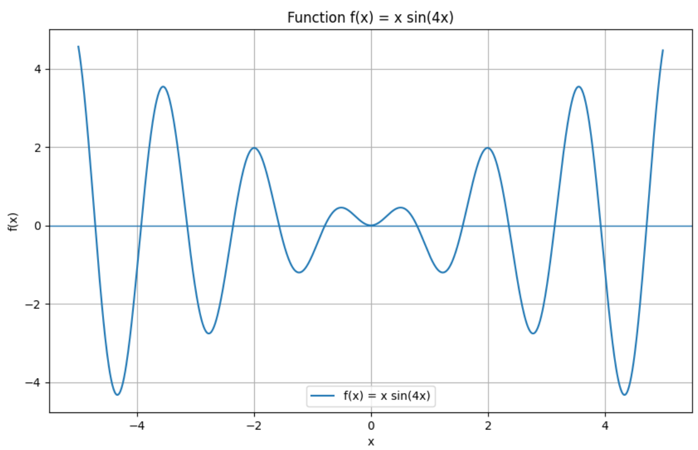
In the code, initial guesses were taken from the interval

$$[-5,5]$$

with a step size of 0.05. For every starting point, the final root and the required number of iterations were stored. This part is related to the first question in the Lab 5 instruction sheet, where the dependence on the starting point is investigated.

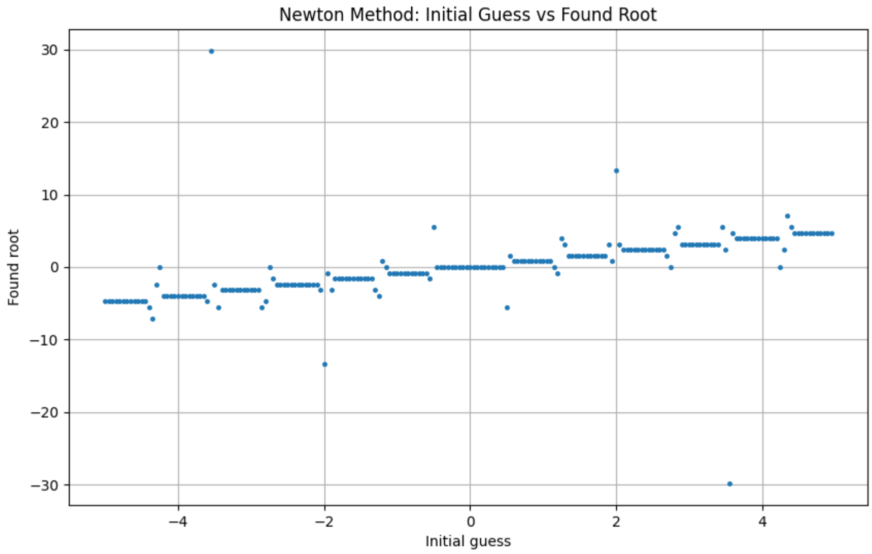
2.1 Function Plot

The first graph shows the shape of the function $f(x) = x\sin(4x)$. It can be seen from the graph that the curve cuts the x-axis many times, so the function has several roots.



2.2 Initial Guess and Found Root

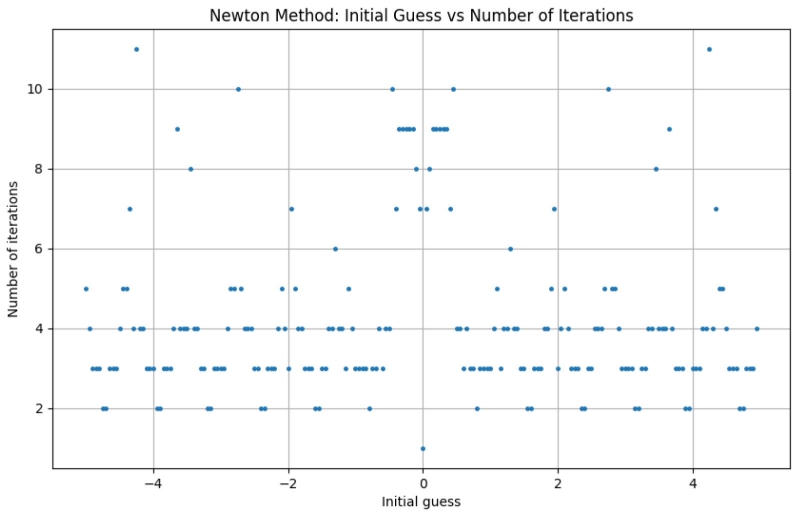
The next plot shows which root is obtained for each initial guess.



The plot is not a smooth curve. This is expected because a small change in the starting point may send Newton's method to another root. Therefore, the result of Newton's method is highly dependent on the initial guess.

2.3 Initial Guess and Number of Iterations

The third plot shows how many iterations are needed for each starting value.



Most of the starting points converge quickly, but some points require more iterations. This shows that the convergence speed is also affected by the starting point.

3-Nonlinear System of Equations

The second part of the lab considers the following system:

$$x^2 + y^2 + z^2 = 2$$
$$x^2 + y^2 = 1$$
$$x^2 = y$$

For Newton's method, the system is rewritten in the form $F(x, y, z) = 0$:

$$F_1(x, y, z) = x^2 + y^2 + z^2 - 2$$
$$F_2(x, y, z) = x^2 + y^2 - 1$$
$$F_3(x, y, z) = x^2 - y$$

So the function vector is:

$$F(x, y, z) = \begin{bmatrix} x^2 + y^2 + z^2 - 2 \\ x^2 + y^2 - 1 \\ x^2 - y \end{bmatrix}$$

4-Jacobian Matrix

To apply Newton's method to a system, the Jacobian matrix is required. The Jacobian consists of the partial derivatives of the function vector.

For this system, the Jacobian matrix is:

$$J(x, y, z) = \begin{bmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \\ 2x & -1 & 0 \end{bmatrix}$$

The Newton update for the system is written as:

$$J(X_n)\Delta X = -F(X_n)$$

and then

$$X_{n+1} = X_n + \Delta X$$

Instead of calculating the inverse of the Jacobian directly, the linear system was solved numerically in the Python code.

5-Numerical Solution of the System

The multivariable Newton method was started from different initial points. One of the starting values used in the program was:

$$X_0 = (0.7,\ 0.6,\ 0.8)$$

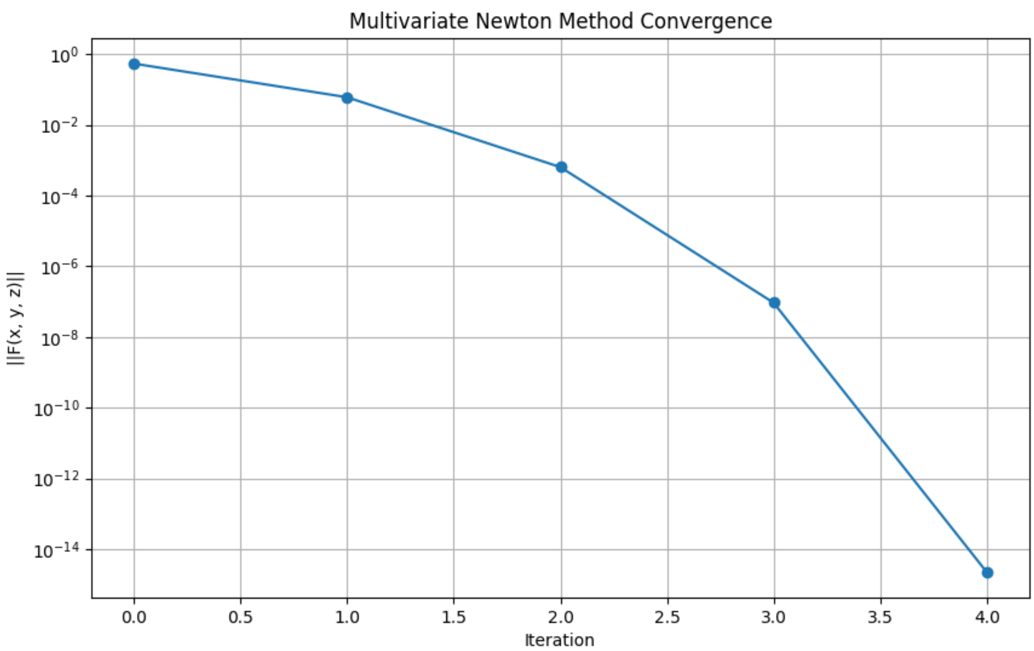
For this initial point, the method converged to approximately:

$$x = 0.7861513778$$
$$y = 0.6180339887$$
$$z = 1.0000000000$$

When this solution is substituted back into the system, the residual becomes almost zero. This means that the solution satisfies the equations accurately.

5.1 Convergence of the Multivariate Newton Method

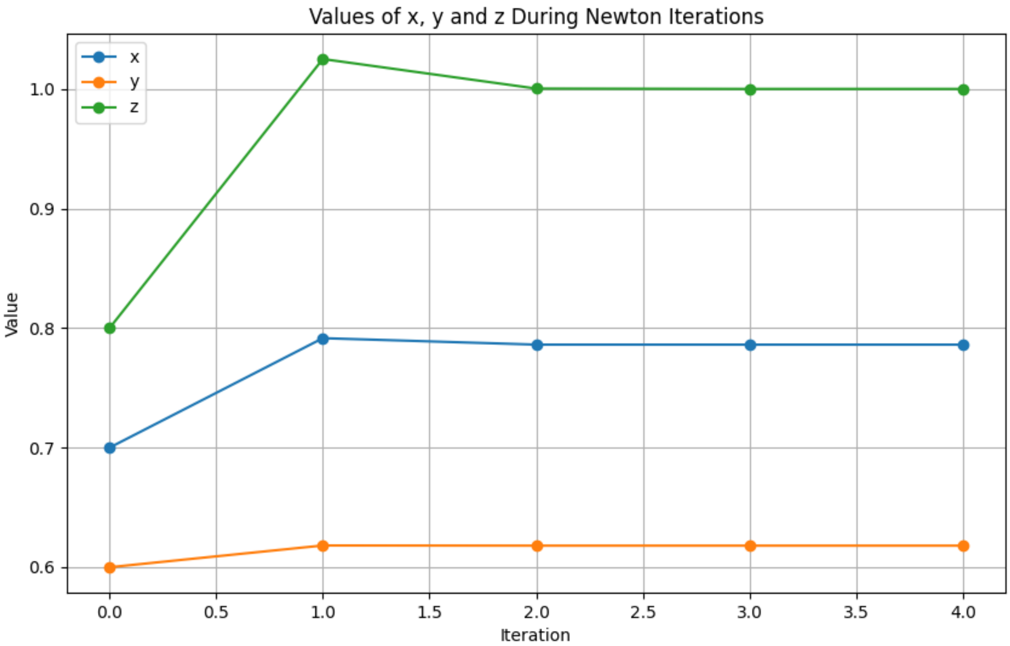
The graph below shows the norm of $F(x, y, z)$during the iterations.



The residual decreases very rapidly. This shows that Newton's method converges in only a few iterations for this starting point.

5.2 Iteration Values of x, y and z

The following graph shows the values of x, y, and $z$at each iteration.



After a few steps, all three variables become nearly constant. This indicates that the iteration reached a stable solution.

6-Aitken Acceleration

The last required part of the lab is the implementation of Aitken's acceleration for a one-dimensional case.

Aitken's method is used to speed up a convergent sequence. In this work, the fixed-point iteration

$$x_{n+1} = \cos(x_n)$$

was used as an example.

The fixed point of this iteration is approximately:

$$x = 0.7390851332$$

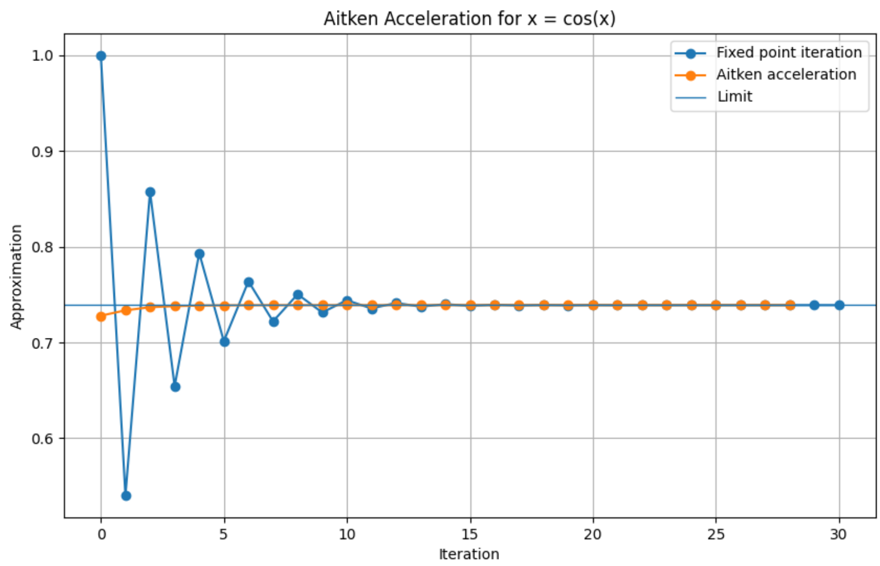
Aitken's formula is:

$$\hat{x}_n = x_n - \frac{(x_{n+1} - x_n)^2}{x_{n+2} - 2x_{n+1} + x_n}$$

This formula uses three consecutive values of the sequence to produce a better approximation.

6.1 Fixed Point Iteration and Aitken Sequence

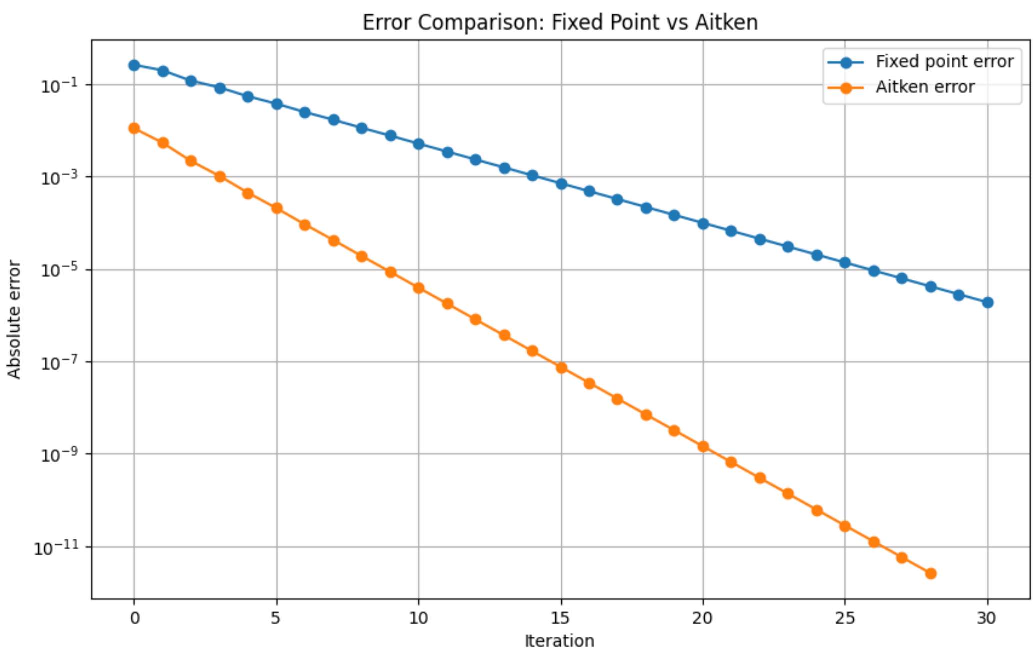
The graph below compares the ordinary fixed-point iteration with the Aitken accelerated sequence.



The ordinary iteration moves toward the limit slowly and oscillates around it. The Aitken sequence reaches values close to the limit much earlier.

6.2 Error Comparison

The next graph compares the absolute errors of the two methods.



The Aitken error decreases faster than the error of the original fixed-point iteration. This confirms that Aitken acceleration improves the convergence rate.

7-Discussion

The results from the first part show that Newton's method can behave differently depending on the initial guess. For a function with many roots, the method may converge to different roots even when the starting values are close to each other.

For the nonlinear system, the Newton method worked effectively. The residual norm decreased quickly, and the values of x, y, and $z$reached stable values after a small number of iterations.

The Aitken acceleration part shows that a slowly converging one-dimensional iteration can be improved. Compared with the ordinary fixed-point iteration, the accelerated sequence approaches the fixed point faster.

8-Conclusion

In this lab, Newton's method was examined for both a single-variable function and a system of nonlinear equations.

The first part showed that the initial guess has a clear effect on the root found by Newton's method and on the number of iterations. The second part showed how Newton's method can be extended to three variables using a Jacobian matrix. The system was solved successfully and the residual decreased to a very small value.

Finally, Aitken's acceleration was applied to a one-dimensional fixed-point iteration. The results showed that Aitken's method improves the convergence speed by producing better approximations from the original sequence.

Overall, Newton's method is a strong numerical technique, but the starting point is important. Acceleration methods such as Aitken's method can also be useful when an iterative method converges slowly.